

Capacity Market Design and Renewable Investment in Spain:

A forensic audit of the *derating factors* of the 2026 capacity mechanism (SA.112483 / RD 932/2024) using hourly ELCC data (2019–2024)

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Abstract

On 29 May 2026 the European Commission approved, under Article 107(3)(c) TFEU, a 9 € billion, ten-year capacity mechanism (CM) for Spain (case SA.112483), in which the system operator remunerates every megawatt *available during a scarcity event* through competitive auctions (European Commission, 2026). The parameter that translates nominal capacity into accreditable megawatts —the *derating factor*— ceases to be an academic abstraction and becomes the variable that allocates 900 € million per year across technologies. This paper audits that parameter. We quantify the *Effective Load Carrying Capability* (ELCC) of six technologies of the Spanish peninsular system using hourly ENTSO-E data (BZN|ES, 2019–2024) and show that three seemingly technical calibration choices create first-order incentive distortions and regulatory arbitrage risks: (i) treating hydro as an aggregate (ELCC = 0.22) under-remunerates reservoir hydro (0.73) and pumped storage (0.64) by a factor of $\sim 3\times$ and over-remunerates run-of-river (0.14), generating *adverse selection* in the auction and destroying the investment signal for storage; (ii) importing northern-European solar *derating factors* (0.01–0.03) ignores the Spanish bimodal profile (ELCC_{solar} = 0.13, summer cooling peaks) and penalises solar *missing money* by $\sim 80\%$; (iii) calibrating gas on the raw history imports a regulatory artifact —the Iberian Exception (2022–2023) inflated CCGT ELCC by +0.072 (+19%)— buying fictitious physical security. Finally, cross-checking with the MIBEL Congestion Monitor (Jaccard ≤ 0.10 between price stress and the top of net demand) confirms that the CM must penalise non-delivery on *physical* adequacy stress, not on *day-ahead* financial volatility. The frame is not “we have computed Spain’s ELCC”: it is *how to calibrate the 2026 CM without distorting investment in storage or paying inflated premiums to gas*.

Keywords: Capacity mechanism, SA.112483, RD 932/2024, ELCC, *derating factors*, adverse selection, *missing money*, regulatory arbitrage, ENTSO-E, MIBEL.

1 Introduction

1.1 Motivation: a 9 € billion mechanism that pays per firm megawatt

The Spanish capacity mechanism has moved from a regulatory proposal to a budgeted transfer. Royal Decree 932/2024 (Ministerio para la Transición Ecológica y el Reto Demográfico, 2024) established its internal architecture, and on 29 May 2026 the European Commission authorised it under State aid rules: a *market-wide* mechanism of 9 € billion over ten years (~ 900 € million/year) in which the transmission system operator (TSO) “remunerates all the capacity needed to meet the reliability standard” and beneficiaries “compete based on the amount of aid requested per MW of capacity available during a scarcity event” (European Commission, 2026). Approval

rests on Article 107(3)(c) TFEU, the CEEAG 2022 Guidelines (European Commission, 2022), and declares alignment with most of the best practices for capacity mechanisms of the *Clean Industrial State Aid Framework* (CISAF) (European Commission, 2025).

The operative phrase —“per MW available during a scarcity event”— is exactly the definition of ELCC. The *derating factor* is the fraction of nominal capacity each technology can accredit as firm before the auction; multiplied by installed capacity and by the clearing price, it sets each project’s capacity revenue. Mis-calibrating it is not an academic rounding error: it mis-allocates 900 € million a year and emits a biased investment signal over a ten-year horizon.

1.2 From descriptive calculation to incentive audit

The ELCC literature is usually framed as a descriptive exercise: measuring each technology’s adequacy contribution. This paper deliberately takes a different stance. We treat each calibration choice of the 2026 CM as an auditable hypothesis and ask: *what incentive distortion or regulatory-arbitrage risk does it introduce?* A well-designed CM, in the Commission’s own words, must not (i) raise prices for consumers, (ii) grant undue advantages to specific operators, nor (iii) hinder cross-border flows (European Commission, 2026). We show that three common calibration shortcuts directly violate criterion (ii) —undue advantages— via mis-specified *derating factors*.

1.3 Research question

What are the empirically justified *derating factors* for the relevant technologies (nuclear, wind, solar, reservoir hydro, run-of-river, pumped storage, gas) in the peninsular system 2019–2024, and what auction distortions does the 2026 CM generate if calibrated with the usual shortcuts (aggregated hydro, imported benchmarks, uncleaned history, price-based activation)? To what extent are the findings robust to the temporal granularity of the data and to the internal split of hydro capacity?

1.4 Contribution

1. **ELCC from real hourly data** (ENTSO-E 16.1.B&C, BZN|ES, 2019–2024) instead of token-free daily generation (REData). For dispatchable technologies with a concentrated temporal profile, the hourly peak capacity factor differs materially from the daily one.
2. **Disaggregation of hydro** into its three sub-technologies (reservoir, run-of-river, pumped storage). The traditional aggregate conceals an adequacy heterogeneity of factor $5\times$, validated through a sensitivity analysis over five capacity-split scenarios.
3. **Quantification of the regulatory bias** of the Iberian Exception (2022–2023) on gas ELCC (+0.072).
4. **Translation into design risks:** we map each statistical finding onto a named auction risk of the 2026 CM (adverse selection, *missing money*, regulatory artifact, price-based activation), with the corresponding calibration recommendation.

2 Literature

Garver (1966) introduced ELCC as a metric of equivalence between capacity additions and demand reductions: a unit with $\text{ELCC}=1.0$ is equivalent, from an adequacy standpoint, to a permanent 1 MW reduction of peak demand. It is this equivalence that the 2026 CM monetises by paying “per MW available in scarcity” (European Commission, 2026).

Mills and Wiser (2013) and Madaeni et al. (2013) formalise the various approaches to computing ELCC under growing renewable penetration, identifying the “saturation” or dilution effect of marginal ELCC as the installed capacity of technologies with correlated profiles increases.

The debate over the need for capacity mechanisms proper —versus an *energy-only* design— is consolidated in Joskow (2008) and Cramton et al. (2013). Batlle and Pérez-Arriaga (2008) discuss specific design criteria for liberalised markets, prefiguring elements of RD 932/2024. The current European framework —Regulation (EU) 2019/943 (European Union, 2019), CEEAG 2022 (European Commission, 2022) and CISAF (European Commission, 2025)— requires the CM to be proportionate and non-discriminatory, which in practice hinges on the quality of the *derating factors*.

The European benchmarks published by National Grid ESO (2024), EirGrid and SONI (2023) and PSE (2023) serve as comparative references, but their transferability to the Spanish case is limited by differences in climatic profile and technology mix —a point that Axis 2 (Section 6.2) turns into a quantified risk.

3 Data

3.1 Generation: ENTSO-E BZN|ES (hourly, 2019–2024)

We download the 72 monthly files of the “Actual Generation per Production Type” dataset (16.1.B&C, area code 10YES-REE—0). The official unit is the interval-average MW (ENTSO-E, 2023): for PT60M (2019-01 to 2022-05) and PT15M (from 2022-06) the conversion to MWh per hour is either direct (PT60M) or requires a moving average (PT15M → PT60M with the mean). Coverage is 99.94% (5 hours missing on 2019-12-24 02–06h UTC, a source gap).

The dataset includes 21 *Production Types*; this work uses seven: nuclear (B14), solar (B16), wind onshore+offshore (B19+B18), reservoir hydro (B12), run-of-river (B11), pumped storage (B10) and *Fossil Gas* (B04).

3.2 Installed capacity: REData (REE)

REData publishes monthly installed capacity by technology (`api.ree.es`, no token). It is forward-filled to hourly resolution to align with ENTSO-E generation. The aggregate “Hydro” capacity (~17 GW) is not broken down into reservoir, run-of-river and pumped storage; the 35/45/20% split is used (Section 6.4), validated by sensitivity.

3.3 Geographic coverage

ENTSO-E BZN|ES corresponds to the peninsular system plus the Balearic Islands; it excludes the Canary Islands. REData generation-by-structure includes the Canaries. The annual difference in wind is roughly −3% (consistent with ~2 TWh/year of Canary wind), absorbed as a documented caveat. The peninsular coverage is the relevant one: SA.112483 opens the mechanism to projects “located in Spain” within the interconnected peninsular system (European Commission, 2026).

3.4 Systemic stress: MIBEL Congestion Monitor

For Axis 4 (Section 6.4) we reuse the 52 530 hours classified by Vilches (2024), of which 1 044 are Orange or Red alerts (1.99% of the total).

4 Methodology

4.1 Naive ELCC

$$\text{ELCC}(\text{tech}, N) = \frac{\overline{\text{gen}}_{\text{tech}}(\mathcal{T}_N)}{\bar{C}_{\text{tech}}(\mathcal{T}_N) \cdot \Delta t} \quad (1)$$

where $\mathcal{T}_N$ are the N records (days in **daily**; hours in **hourly**) with the highest net demand in the period, $\bar{C}$ is the mean installed capacity over those records, and Δt is the interval duration (24h or 1h). Net demand: $D^{net} = \sum_i \text{gen}_i - \text{gen}_{\text{wind}} - \text{gen}_{\text{solar}}$.

For **daily** we use $N = 100$ (top-100 days, matching the convention of [ENTSO-E \(2023\)](#)); for **hourly**, $N = 2\,400$ (top-2 400 hours, the same temporal volume).

4.2 Marginal ELCC

Following [Mills and Wiser \(2013\)](#) it is approximated by a finite difference:

$$\text{marginal_ELCC}(\text{tech}) = \frac{\text{ELCC}(C + \Delta C) - \text{ELCC}(C)}{\Delta C} \quad (2)$$

with $\Delta C = 1$ GW. The operational assumption is that generation in peak hours does not increase when capacity is added (dispatch saturation); this provides an upper bound on the dilution from additional capacity. By construction, all marginals are negative.

4.3 Market regimes

The period is segmented into three regimes:

- *Pre-crisis*: 2019-01-01 to 2021-12-31.
- *Iberian Exception*: 2022-01-01 to 2023-12-31 (gas adjustment mechanism, RDL 10/2022).
- *Post-exception*: 2024-01-01 onwards (return to the ordinary regime with high renewable penetration).

The *regulatory bias* of a technology is defined as the difference between its ELCC over the full period and its ELCC excluding the Iberian Exception. Operationally, it is the fictitious firmness premium the 2026 CM would “buy” if calibrated on the raw history.

5 Results

5.1 ELCC by technology: *daily* vs *hourly*

Table 1 summarises the ELCC obtained with both granularities, restricted to the period without the Iberian Exception. For nuclear, solar and wind the values converge ($|\Delta| < 0.01$), confirming methodological robustness. The significant differences appear in hydro and gas —precisely the technologies on which the 2026 CM calibration risk falls.

Table 1: ELCC by technology, *daily* (REData) vs *hourly* (ENTSO-E), without the Iberian Exception, N equivalent to 100 days.

Technology	ELCC <i>daily</i>	ELCC <i>hourly</i>	Δ
Nuclear	0.927	0.919	−0.008
Solar (PV+T)	0.126	0.126	+0.000
Wind	0.152	0.145	−0.007
Total hydro	0.222	0.334	+0.112
Reservoir hydro	—	0.728	new
Run-of-river hydro	—	0.139	new
Pumped hydro	—	0.642	new
CCGT (<i>daily</i>) / Fossil Gas (<i>hourly</i>)	0.372	0.457	+0.085

5.2 Finding: disaggregated hydro

Table 1 contains the paper’s main finding. The “Hydro = 0.22” aggregate of the daily approach is an opaque average of three very different technologies:

- **Reservoir** (~35% capacity): ELCC = 0.728. Almost as firm as nuclear.
- **Pumped storage** (~20% capacity): ELCC = 0.642. Dispatchable, optimised for peak hours.
- **Run-of-river** (~45% capacity): ELCC = 0.139. Low, as expected (non-dispatchable).

The economic interpretation of this dispersion —adverse selection in the auction— is developed in Axis 1 (Section 6.1). Figure 1 shows ELCC by threshold N for all hourly technologies.

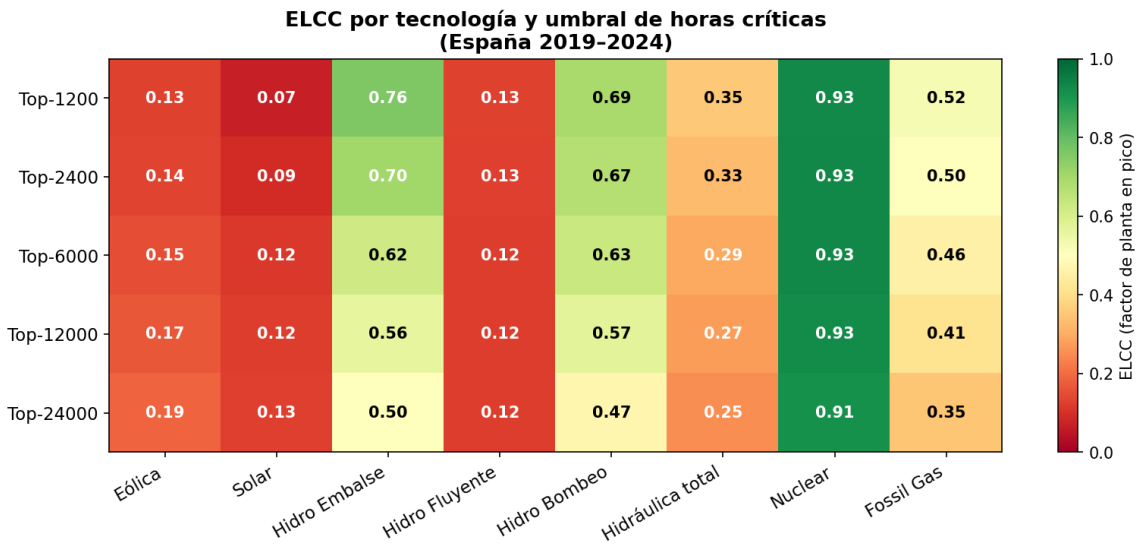


Figure 1: Hourly ELCC by technology and threshold N . The “Reservoir hydro” column is virtually flat around 0.5–0.75 across the various N , indicating high structural firmness.

5.3 Regulatory bias: the Iberian Exception

Table 2 compares ELCC with and without the Iberian Exception. For strict CCGT (*daily*, REData) the bias is +0.072 (+19% relative to 0.372). For total Fossil Gas (*hourly*, ENTSO-E, includes gas cogeneration) the bias is smaller, +0.041, because the intervention mainly affected the merit-order CCGT and not the non-intervened industrial cogeneration. The calibration implication is developed in Axis 3 (Section 6.3).

5.4 Marginal ELCC

Table 3 reports the marginals for $\Delta C = 1$ GW in *hourly*. All are negative (saturation). The most pronounced correspond to technologies with the smallest base installed capacity (pumped storage ~3.4 GW, reservoir ~6 GW, nuclear ~7 GW), where an additional GW represents a proportionally large expansion.

5.5 Sensitivity to HYDRO_CAP_RATIOS

REData’s aggregate hydro capacity (~17 GW) is split into three sub-technologies with approximate ratios (35/45/20%) based on public REE statistics. Table 4 shows ELCC under five alternative scenarios with the aggregate capacity held fixed.

Table 2: Regulatory bias by technology (bias = $\text{ELCC}_{2019-24} - \text{ELCC}_{\text{without Iberian Exception}}$).

Technology	Bias <i>daily</i> (REData)	Bias <i>hourly</i> (ENTSO-E)
Nuclear	+0.003	+0.011
Wind	−0.024	−0.009
Solar	+0.014	−0.034
Total hydro	−0.025	−0.005
Reservoir	—	−0.024
Run-of-river	—	−0.010
Pumped	—	+0.027
CCGT / Fossil Gas	+0.072	+0.041

Table 3: Marginal ELCC, $\Delta C = 1$ GW, *hourly*, without the Iberian Exception.

Technology	Base ELCC	ELCC +1 GW	Marginal	Base cap. (GW)
Wind	0.145	0.140	−0.0053	28.3
Solar	0.126	0.118	−0.0082	18.4
Run-of-river	0.139	0.123	−0.0160	7.7
Pumped storage	0.642	0.497	−0.1453	3.4
Reservoir	0.728	0.624	−0.1042	6.0
Nuclear	0.919	0.806	−0.1132	7.1
Fossil Gas	0.457	0.441	−0.0161	27.4

Robustness of the finding: in the most adverse scenario for reservoir (*Generous-reservoir*, 40% cap), ELCC remains $0.637 > 0.60$. In the worst case for pumped storage (*Generous-pumped*, 25% cap), ELCC falls to 0.514, still clearly firm. The qualitative conclusion “reservoir and pumped storage are firm technologies” holds across all five scenarios —the adverse-selection risk of Axis 1 does not depend on the exact split.

5.6 Comparison with European benchmarks

Table 5 compares Spain with the UK, Ireland and Poland. The peninsular solar ELCC (~ 0.13 *hourly*) exceeds the UK and Irish values by 4–13 $\times$. The reason is Spain’s bimodal demand profile: critical hours are not only winter-nocturnal but also summer-diurnal (cooling under high irradiance). Figure 2 quantifies this bimodality, the basis of Axis 2 (Section 6.2).

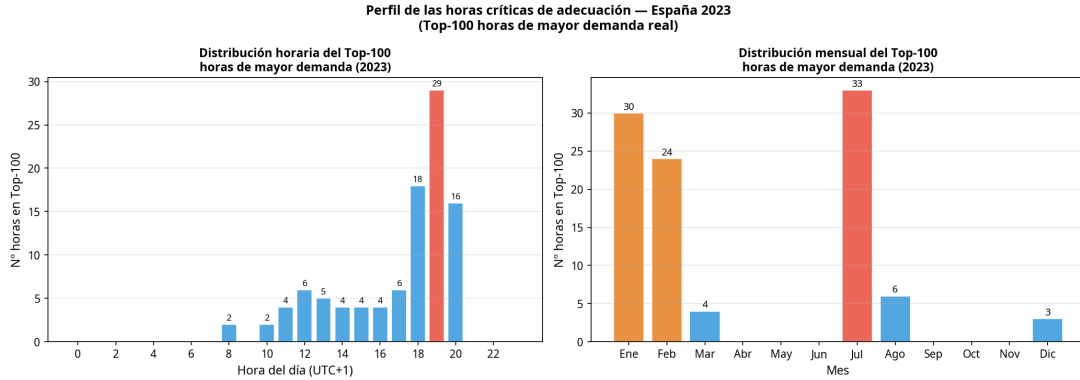


Figure 2: Hourly and monthly profile of the top-100 highest-demand hours (Spain 2023). Double peak: winter (18–20h, Jan/Dec) and summer (12–17h, Jul/Aug). $\sim 50\%$ of the critical hours have sun available.

Table 4: Sensitivity of sub-hydro ELCC to HYDRO_CAP_RATIOS (top-2 400 hours, without the Iberian Exception).

Scenario	Reservoir / RoR / Pumped	Reservoir	RoR	Pumped
Base	35 / 45 / 20	0.728	0.139	0.642
Conservative-reservoir	30 / 50 / 20	0.849	0.125	0.642
Generous-reservoir	40 / 40 / 20	0.637	0.156	0.642
Conservative-pumped	35 / 50 / 15	0.728	0.125	0.856
Generous-pumped	35 / 40 / 25	0.728	0.156	0.514
Observed range		[0.64, 0.85]	[0.13, 0.16]	[0.51, 0.86]

Table 5: Spain ELCC (*hourly*, without the Iberian Exception) vs published European benchmarks.

Country / System	Wind	Solar	Nuclear	Hydro
Spain (this work)	0.15	0.13	0.92	0.33[†]
UK (National Grid ESO, 2024)	0.08	0.03	0.92	0.15
Ireland (EirGrid and SONI, 2023)	0.10	0.01	—	0.20
Poland (PSE, 2023)	0.08	0.02	—	0.10

[†] Total hydro (Reservoir 0.73, RoR 0.14, Pumped 0.64).

5.7 Systemic stress vs adequacy: cross-check with MIBEL

The 1 044 O/R hours classified by the MIBEL Congestion Monitor ([Vilches, 2024](#)) overlap with the top- N highest-net-demand hours only marginally: a Jaccard index between 0.05 (top-50) and 0.10 (top-1 000). This confirms that the stress detected by MIBEL —structural tension on the ES-FR interconnector and extreme prices— is economic in nature, not physical adequacy. Figure 3 visualises this decoupling, the core of Axis 4 (Section 6.4).

Tabla 6.1: Solapamiento entre definiciones de horas críticas

N (top net demand)	Hours O/R alerts	Hours top-N	Overlap	Jaccard
50.0	1044.0	50.0	18.0	0.051
100.0	1044.0	100.0	29.0	0.074
250.0	1044.0	250.0	40.0	0.075
500.0	1044.0	500.0	64.0	0.084
1000.0	1044.0	1000.0	117.0	0.097

Figure 3: Overlap between MIBEL O/R alerts and the top- N highest-net-demand hours. Jaccard ≤ 0.10 at every threshold: economic stress is not a proxy for physical insufficiency.

6 Design audit: four calibration risks of the 2026 CM

This section translates the results of Section 5 into design risks for the mechanism approved in SA.112483. Each axis follows the same structure: the *link* to the auction mechanics, the *forensic argument* on the distortion introduced by shortcut calibration, and the *recommendation* for correct calibration.

6.1 Axis 1 — Aggregated hydro: adverse selection and under-remuneration of storage

Link. The auction awards a price per firm MW to each plant. The *derating factor* sets how many firm MW each unit accredits per MW installed.

Forensic argument. If the regulator indexes the *derating factors* on “total hydro” (ELCC = 0.22), the mechanism is born economically broken. It would be *under-remunerating* reservoir (0.73) and pumped storage (0.64) by a factor close to $3\times$, and *over-remunerating* run-of-river (0.14) by almost double. The result is a classic *adverse-selection* problem: run-of-river—which, depending on instantaneous flow, cannot guarantee power in an hour of physical stress—receives a firmness premium it does not deserve, while reservoir and pumped storage—the hydro storage that does deliver power in scarcity—are underpaid. The long-term investment signal for storage, which SA.112483 states it wants to foster to reach the national flexibility target (European Commission, 2026), would be destroyed in the very first auction parameter. The sensitivity of Section 5.5 confirms that reservoir stays firm ($\in [0.64, 0.85]$) and pumped storage too ($\in [0.51, 0.86]$) across the five scenarios: the risk is not an artifact of the assumed capacity split.

Recommendation. Disaggregate hydro into differentiated auctions or, at a minimum, calibrate *derating factors* specific to each production-unit code (ESIOS), with values around 0.73 (reservoir), 0.64 (pumped) and 0.14 (run-of-river). This is the only way to mitigate adverse selection and preserve the storage investment signal that CISAF itself prioritises.

6.2 Axis 2 — Bimodal profile: solar *missing money*

Link. The solar *derating factor* determines what fraction of solar *missing money* the plant recovers via capacity revenue.

Forensic argument. The CM design in northern Europe (UK, Poland, Ireland) assumes a virtually nil solar adequacy value (1–3%) because there critical hours occur strictly on winter nights. But the Spanish profile is *bimodal* (Figure 2): the top of critical hours includes a massive summer peak (July/August, 12–17h) driven by cooling. With $ELCC_{\text{solar}} = 0.13$, peninsular PV physically coincides with ~50% of the system’s critical adequacy hours. If the 2026 CM blindly imports the National Grid or PSE parameters, it would penalise solar *missing money* by ~80% (from 0.13 to ~0.03), a de facto discrimination that the non-discrimination criterion of SA.112483 (European Commission, 2026) does not allow.

Recommendation. Calibrate the solar *derating factor* on the domestic bimodal profile (~0.13), not on imported benchmarks. The Spanish system needs to recognise the PV contribution to covering summer thermal peaks.

6.3 Axis 3 — Decontaminating the Iberian Exception in gas

Link. Combined-cycle plants (CCGT) will be the key back-up actors in the first CM auctions.

Forensic argument. Using the raw five-year history to set the gas *derating factor* introduces a severe bias. During the Iberian Exception (2022–2023), the adjustment mechanism artificially altered the CCGT merit order, forcing high dispatch to export subsidised energy to France. The finding that this inflated strict CCGT ELCC by +0.072 (+19%, Table 2) demonstrates that this firmness increase was neither physical nor climatic, but a *temporary regulatory artifact* of an economic incentive that no longer exists.

Recommendation. Clean 2022–2023 from the history before calibrating gas firmness rights for the 2026 auction, in line with the ENTSO-E doctrine of excluding intervention periods (ENTSO-E, 2023). Otherwise, the system would “buy” a false sense of physical security and pay an inflated premium to gas—exactly the “undue advantage to specific operators” that SA.112483 requires to be avoided (European Commission, 2026).

6.4 Axis 4 — Jaccard decoupling: the CM is not a price insurance

Link. There is a design tension over whether the CM should measure its stress —and trigger its non-delivery penalties— from the *spot* price or from physical variables.

Forensic argument. The cross-check with the MIBEL Congestion Monitor is conclusive: a Jaccard index ≤ 0.10 between economic-stress alerts and the top of net demand (Figure 3) shows that price stress —typically caused by congestion rents on the interconnector with France or gas price spikes— *does not correlate* with the physical insufficiency of internal generation. A CM that activated penalties on price criteria would be remunerating and penalising the wrong event.

Recommendation. Validate the separate design of RD 932/2024: the CM should be a complement to the *spot* market, indexing its non-delivery penalties strictly to hours of *physical* adequacy stress, not to spikes of *day-ahead* financial volatility. Activating penalties on price would distort plants’ risk hedging and inflate bid premiums —raising the cost of the mechanism for the consumer, against the Commission’s first criterion (European Commission, 2026).

6.5 Marginal ELCC: do not over-remunerate redundant capacity

Cutting across the four axes, Table 3 shows that all marginals are negative (dilution from saturation). Auctions should incorporate marginal-cost calculation methodologies so as not to over-remunerate additions in already saturated technologies that bring no adequacy value at the margin.

7 Limitations

7.1 Geographic coverage

ENTSO-E BZN|ES excludes the Canaries. The ~2 TWh/year of Canary wind is not reflected, producing a downward bias of ~3% in wind ELCC. As the Canaries are an isolated system with its own dispatch mechanism, this exclusion is defensible for the peninsular CM.

7.2 Fossil Gas definition

ENTSO-E “Fossil Gas” includes natural-gas-fired cogeneration, not only CCGT. This is functionally correct for measuring the system’s total firm gas, but it is not 1:1 with the “Combined Cycle” category that the CM treats as a separate auction. Axis 3 therefore uses strict CCGT (REData) to quantify the bias.

7.3 Internal hydro split

REData publishes aggregate hydro installed capacity (~17 GW) without breaking down reservoir, run-of-river and pumped storage. The 35/45/20% split is assumed, based on public REE statistics. The sensitivity (Section 5.5, Table 4) confirms that the qualitative finding is robust, but precise numerical calibration would require access to ENTSO-E 14.1.A (Installed Capacity per Production Type), pending.

7.4 Plant-level granularity

ESIOS (api.esios.ree.es) would publish hourly generation per unit (not aggregated), enabling plant-level ELCC analysis —exactly the granularity that Axis 1 recommends for the hydro *derating factors*. The token request is pending; the refinement is future work.

7.5 Short post-exception period

Only 2024 is available as a post-Exception observation. The findings on the post-exception regime (especially expanded solar and pumped storage) should be revisited with 2025–2026 data, ideally before the mechanism’s first auction.

8 Conclusions

1. For nuclear, solar and wind, hourly granularity does not materially change ELCC relative to daily ($|\Delta| < 0.01$). The daily REData approach is methodologically valid for these technologies.
2. **Axis 1.** The “Hydro = 0.22” aggregate conceals three very different regimes (**reservoir 0.73, run-of-river 0.14, pumped 0.64**). Calibrating the CM on the aggregate generates adverse selection: it underpays storage $\sim 3\times$ and overpays run-of-river. The finding is robust to five capacity-split scenarios (reservoir [0.64, 0.85], pumped [0.51, 0.86]).
3. **Axis 2.** The peninsular solar ELCC (0.13) is $4\text{--}13\times$ the northern-European benchmarks because of the bimodal profile. Importing foreign parameters would penalise solar *missing money* by $\sim 80\%$.
4. **Axis 3.** The Iberian Exception inflated strict CCGT ELCC by +0.072. The 2026 CM must exclude 2022–2023 from gas calibration so as not to pay a fictitious firmness premium.
5. **Axis 4.** MIBEL stress is economic (Jaccard ≤ 0.10 with physical adequacy). The CM should index penalties on physical stress, not on the *spot* price; the separate design of RD 932/2024 is consistent with this evidence.
6. **Joint reading.** The three undue advantages that State aid rules require to be avoided—higher prices, advantage to specific operators, distortion of trade—materialise, in the Spanish case, through mis-calibrated *derating factors*. Getting SA.112483 right is not “computing ELCC”: it is calibrating the mechanism without distorting investment in storage or paying inflated premiums to gas.

Reproducibility The full code, data and figures are available at <https://github.com/cmvddata/capacity>. Results are reproducible via `python main.py -freq hourly`; the unit tests (`pytest tests/`, 27/27 *passing*) cover the critical methodological decisions.

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